Country Fair VR: Rule-Based Pattern Detection and Facial-Expression Emotion Recognition for Personalized Stroke Rehabilitation

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Abstract

Stroke rehabilitation requires intensive, long-term interventions tailored to each survivor, yet most VR-based systems still rely on standardized protocols and static environments. This paper presents *Country Fair VR*, a virtual reality serious game prototype for the Meta Quest 3, integrating pattern recognition at two complementary levels: a **hybrid personalization module**, which automatically adapts mini-game difficulty through a rule based algorithm recognizing patterns of success and failure and an **emotion recognition module**, which captures a player's facial expression, classifies it into a valence based category and relays it to a peer's mini-game environment to mitigate social isolation. The two modules were evaluated in separate user studies with non clinical participants (N=38 and N=17, respectively). The personalization module achieved a mean system usability scale (SUS) score of **82/100** ("Excellent"), with **87%** of participants agreeing the difficulty adapted to their specific needs. The emotion recognition module's facial expression based representation scored highest in social emotional connection among three tested formats.

1 Introduction

Stroke is a leading cause of long-term disability, eroding survivors' motor function, cognition and quality of life [2]. Recovery depends on intensive, sustained rehabilitation to stimulate neuroplasticity [5], yet conventional programs rely on repetitive, standardized exercises that undermine motivation. Virtual Reality (VR) enables interactive, goal-oriented environments [5], but most VR-based systems lack personalization: they track neither in-session performance nor the player's affective state, both pattern-rich signals [6].

This paper presents *Country Fair VR*, a personalized VR serious game for post-stroke rehabilitation, as a case of applied pattern recognition at two levels: (i) rule-based detection of performance patterns (success/failure) driving real-time difficulty adaptation through a **Hybrid Personalization Module** and (ii) facial-expression recognition of affective patterns, conveyed by an **Emotion Recognition Module**. Unlike [3], which proposed the concept, we report the implemented DDA algorithm and its usability evaluation (N=38) and integrate within the same game an emotion module developed in parallel by our group.

2 Country Fair VR

Country Fair VR is a VR serious game prototype for the Meta Quest 3, composed of a Game module, a Personalization module and an Emotion Recognition module (Figure 1) and featuring two therapeutically grounded mini-games set in a virtual country fair [3]. Upon entering the fair, the player receives a brief introduction before the tents housing each mini-game become available for selection, within each mini-game, reaching a target score returns the player to the fair with a congratulatory message, after which any mini-game can be replayed.

In the **Archery** mini-game, the player uses one of their hands to pull the bowstring and shoot balloons of a specific color, following a dual task color matching pattern that targets bi manual strength, proprioceptive training and selective attention. In the **Frisbee** mini-game, the player grabs and throws a frisbee toward specific target areas so that a dog can retrieve it and return it, targeting elbow extension, wrist coordination, visuospatial integration and reaction capability.

2.1 Personalization Module

The Personalization Module implements a rule-based Dynamic Difficulty Adjustment (DDA) system, embodied by the character *Carny Wise* (Figure 1), who explains the mini-game rules and notifies the player (dialogue and sound cue) whenever difficulty changes. Rather than a learned model, it monitors the stream of task outcomes against fixed thresholds to detect recognizable performance patterns (*e.g.* sustained failure indicating frustration, sustained success indicating under-challenge), keeping the player in a "Flow State" [1]. The algorithm (Figure 2) has two branches. *Failure Handling* decreases difficulty after M consecutive misses (M=[4]). *Precision Analysis* runs on each score: precision is $p = 1/n$, where n is the number of attempts needed to score, so $p \geq 0.6$ only when n=1 and $p \leq 0.4$ when $n \geq 3$ (n=2 is neutral). High precision increments an *Excel* counter and low precision a *Struggle* counter, each decrementing the other, reaching T_E=[3] or T_S=[3] triggers a difficulty increase or decrease, respectively. These values are provisional, not clinically calibrated. The module is hybrid: a therapist can override difficulty at any time via keyboard, a preference reported by 70% of surveyed healthcare professionals [4].

2.2 Facial-Expression Pattern Recognition

As a complementary component, an Emotion Recognition Module captures the player's facial expressions through a webcam and classifies them into valence-based categories (positive, neutral, negative). The recognized emotional state is relayed to a peer's VR mini-game environment, mitigating the social isolation typical of individual rehabilitation sessions. Three representation formats were implemented: an emoji-based cue, a continuous positive to negative slider and a facial expression reference image, selectable randomly or manually chosen. As the Meta Quest 3 has no internal face-tracking cameras, in the tests the expressions were captured by a laptop webcam and streamed to the game.

3 Evaluation

Two descriptive studies with non-clinical adults assessed each module. The personalization study (N=38) evaluated usability and whether players noticed the adaptation to their performance: the prototype obtained a mean **SUS of 82/100 ("Excellent")** and **82% (31/38)** noticed the game adapting, with **87% (33/38)** agreeing it adapted to their abilities. The emotion study (N=17) compared emoji, slider and facial-expression formats, the latter obtaining the highest medians for empathy and social connection (6, against 4–5). Emotions were simulated rather than spontaneously recognized, covering the seven supported expressions (neutral, happy, surprise, sad, angry, fear and disgust), which map to positive (happy, surprise), negative (sad, angry, fear, disgust), or neutral valence. Results reflect perceived adaptation, without a non-adaptive control, clinical validation is pending.

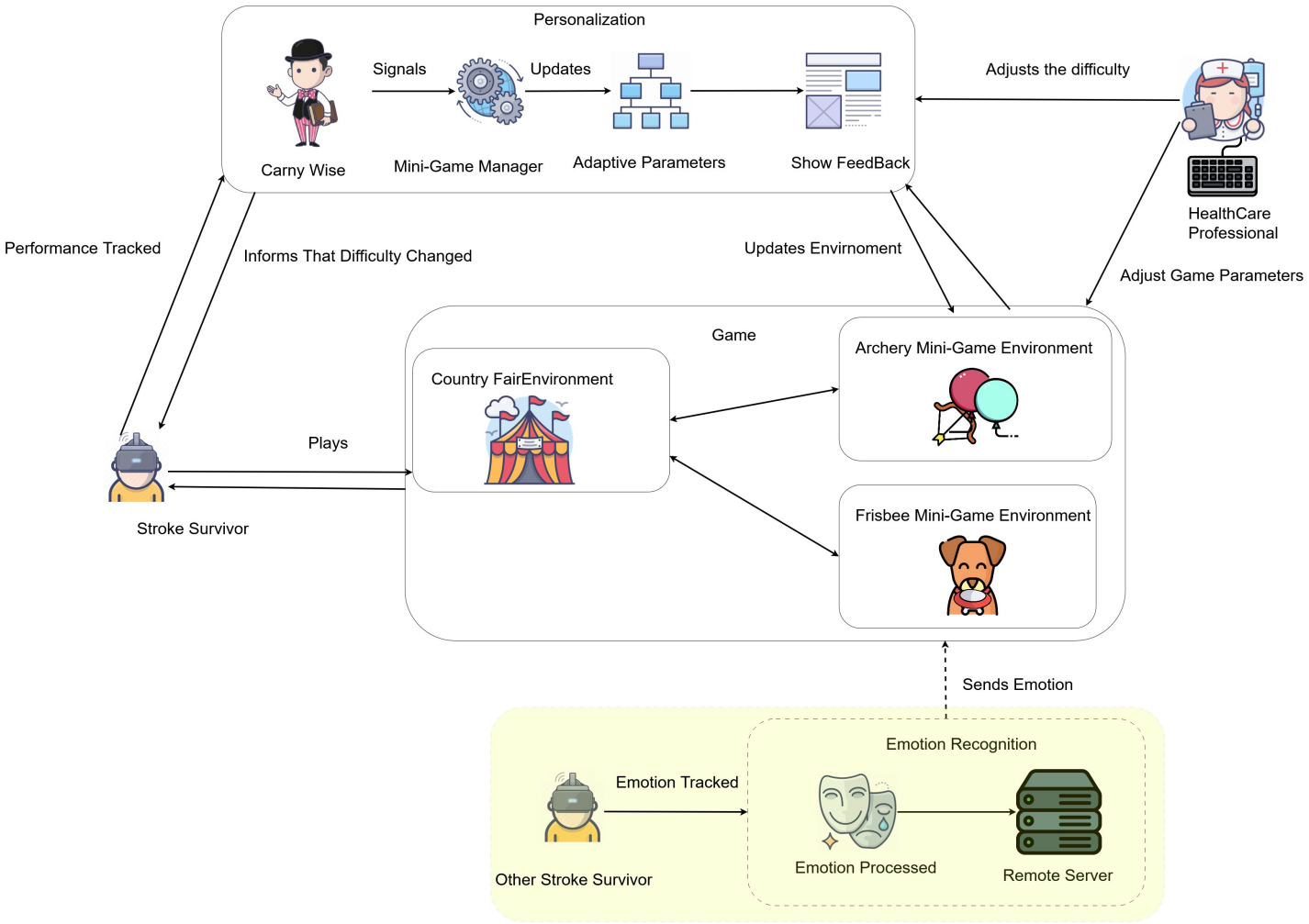


Figure 1: Architecture of *Country Fair VR*, showing the Personalization module (rule based DDA and therapist override), the Game module (Country Fair environment and mini games) and the Emotion Recognition module (facial expression capture and relay to a peer).

```
on attempt:
  if failed:
    miss++
    if miss >= M:
      decrease_difficulty(); miss = 0
  else:                      // scored
    n = miss + 1; miss = 0 // attempts to score
    p = 1/n
    if p >= 0.60:
      excel++; struggle = max(struggle-1, 0)
    if p <= 0.40:
      struggle++; excel = max(excel-1, 0)
    if excel >= T_E:
      increase_difficulty(); excel = 0
    if struggle >= T_S:
      decrease_difficulty(); struggle = 0
// therapist may override difficulty anytime
```

Figure 2: Simplified pseudocode of the rule-based DDA algorithm.

4 Final Remarks and Next Steps

Country Fair VR illustrates two complementary forms of pattern recognition in VR-based stroke rehabilitation: rule-based detection of success/failure patterns driving real-time adaptation and facial-expression recognition of affective patterns supporting social connectedness. Preliminary results with non-clinical participants were encouraging: the prototype achieved an excellent usability score **(SUS 82/100),** most participants noticed and agreed with the difficulty adaptation **(82% and 87%)** and the facial-expression representation stood out in social-emotional connection. These findings, however, reflect perceived adaptation and simulated emotions, the two modules were evaluated separately and the accuracy of the emotion recognition was not assessed. Future work includes extending the Personalization Module toward learned, data-driven detection from longitudinal player data and clinical validation at a rehabilitation center with healthcare professionals.

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